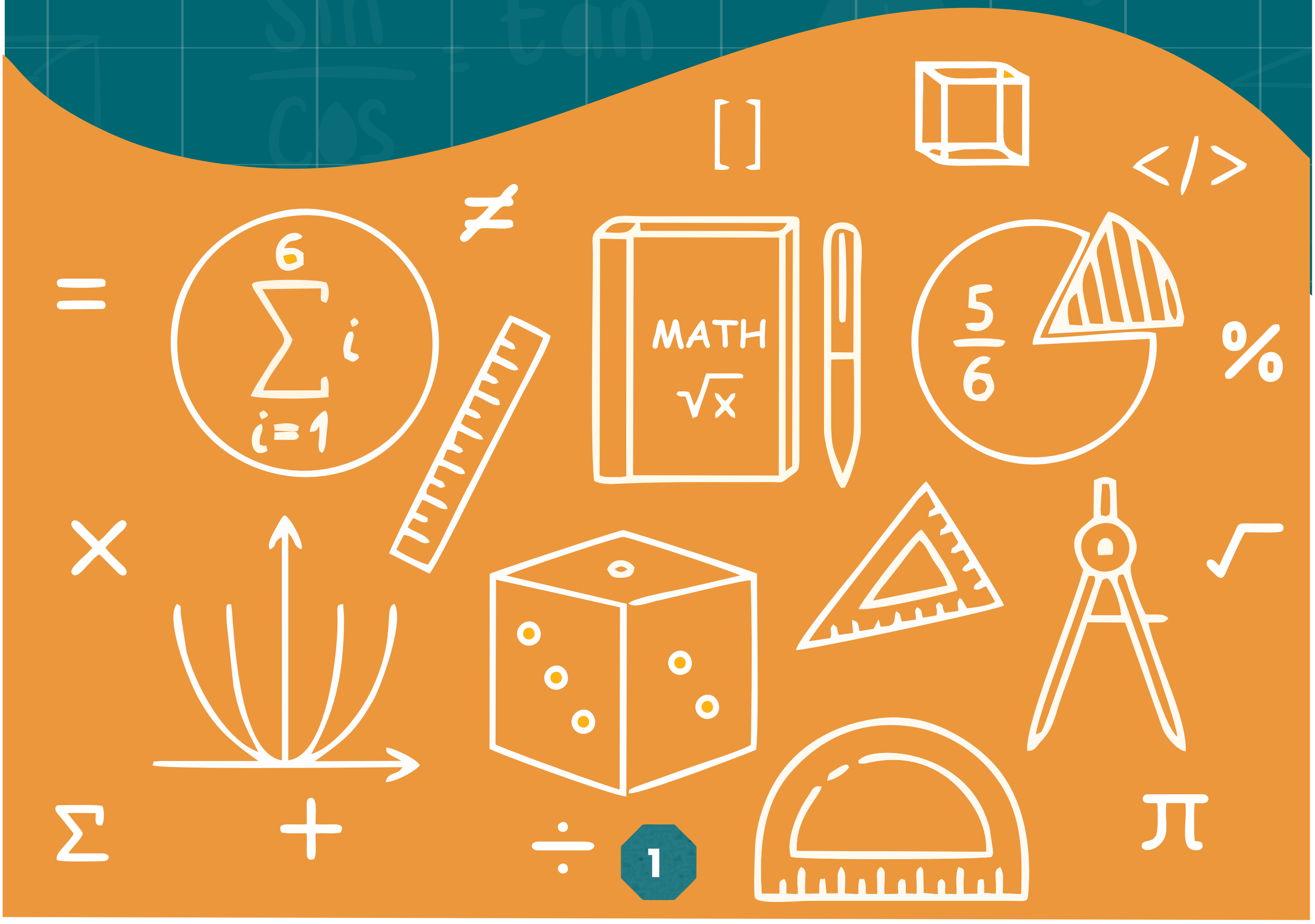
UNIT 1

SETS



SETS

SET

In mathematics collection of things is called set.

A set is a collection of 'well defined' and 'distinct' objects.

'Well defined' means a set must have some specific property.
'Distinct' means different objects.
'Object' is known as an element or a member that can represent anything (numbers, letters, people, etc.).

SET-BUILDER NOTATION

We have already discussed tabular and descriptive forms of set in previous classes. Here we study the set builder notation in detail.

Set-builder notation defines a set by describing the properties or rules that its elements must satisfy, rather than listing them. It is written as $\{ x \mid \text{property of } x \}$, where x is an element of the set, and | means such that.

Example: B = {x | x is an even number greater than 2}.This describes the set of even numbers greater than 2.

Set-builder notation also uses other symbols like $\in$, $\notin$, $\wedge$ and $\vee$. $\in$ means belongs to, $\notin$ means does not belong to, $\wedge$ means AND and $\vee$ means OR.

Example: $C = \{x \mid x \in N \wedge x < 9\}$. This means a Set C whose elements belong to the set of Natural Numbers (N) and are less than 9. In the tabular form the set will be written as C = {1, 2, 3, 4, 5, 6, 7, 8}

TYPES OF SET

1. Empty Set ({}): It is an empty set with no elements.
Example: {}.

2. Singleton Set ({a}): It is a set that has only one element.
Example: {2}.

3. Finite Set ({a, b, c}): A finite set has a countable number of elements.
Example: {1, 2, 3, 4, 5}.

4. Infinite Set (Natural Numbers): An infinite set has an unlimited number of elements that continue forever.
Example: {x | x is a natural number}

SET OPERATIONS

1. Union (∪): The set of all elements that are in either or both sets.
Example: If $A = \{1, 2\}$ and $B = \{2, 3\}$, then $A \cup B = \{1, 2, 3\}$.

2. Intersection (∩): The set of elements common to both sets.
Example: If $A = \{1, 2\}$ and $B = \{2, 3\}$, then $A \cap B = \{2\}$.

3. Difference (-): The set of elements in one set that are not in the other.
Example: If $A = \{1, 2, 3\}$ and $B = \{2, 3\}$, then $A - B = \{1\}$.

4. Complement ('): The set of elements not in the given set, within a universal set.
Example: If $U = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$, then $A' = \{3, 4\}$.

5. Symmetric difference (Δ): The symmetric difference of two sets, often denoted as $A \Delta B$, is the set of elements that are in either of the sets A or B, but not in both. The symmetric difference of two sets

$$A \Delta B = (A - B) \cup (B - A)$$

Key properties of symmetric difference:
1. Commutative: $A \Delta B = B \Delta A$
2. Associative: $(A \Delta B) \Delta C = A \Delta (B \Delta C)$
3. Identify element: The symmetric difference of any set A with the empty set $\varnothing$ is A.

$$A \Delta \varnothing = A$$

4. Self-inverse: A set symmetric-differenced with itself is the empty set:

$$A \Delta A = \varnothing$$

EXERCISE 1.1

1. Write the following sets in tabular form.
(i) A = Set of first five natural numbers
(ii) B = Set of integers between -2 and 3
(iii) C = Set of prime numbers between 10 and 30

2. From the sets given below identify:

(i) Finite sets **(ii)** Infinite sets

A = {1,2,3,composite numbers between 10 and 30}. **B =** {1,3,5,7,.....}.

C = Set of all multiples of 10, E= $\{x \mid x \in Z\ x<0\}$

D = Set of {4,5}

3. If A={1, 2, 3, 4, 5, 6}, B={2, 3, 5, 7, 11} and C={1, 3, 5, 7, 9}, Find

(i) $A \Delta B$ **(ii)** $B \Delta C$

(iii) $C \Delta A$

4. If A = {1,2,3,4,5,6} and U={1,2,3,4,5,6,7,8,9,10,11,12} then verify with examples:

(i) $A \cap A' = \varnothing$ **(ii)** $A \cup A' = U$

Recognize Some Important Sets:

Some important sets of numbers along their notations are given as:

- **Set of Natural Numbers** N = {1, 2, 3, ...}.
- **Set of Whole Numbers** W = {0, 1, 2, 3, ...}.
- **Set of Integers** Z = {..., −3, −2, −1, 0, 1, 2, 3, ...}.
- **Set of Rational Numbers** $Q = \{x \mid x = p/q,\ p \text{ and } q \in Z,\ q \neq 0\}$.
- **Set of Even Numbers** E = {..., −6, −4, −2, 0, 2, 4, 6, ...}.
- **Set of Odd Numbers** O = {..., −5, −3, −1, 1, 3, 5, ...}.
- **Set of Prime Numbers** P = {2, 3, 5, 7, ...}.

Subset:

Set X is said to be a subset of a set Y if every element of X is also an element of Y. Symbolically, we write as $X \subseteq Y$.

Example: Let X = {1,2,3} and Y = {1,2,3,4,5}. We see that $X \subseteq Y$ ..

(i) Empty set is a subset of every set.

(ii) Every set is a subset of itself.

Define Proper subset ($\subset$) and an Improper subset ($\subseteq$):

1. Proper subsets:

Any subset of a set A is said to be a proper subset of A, if it is not equal to A. If B is a proper subset of A, then we write it as $B \subset A$.

Example 1: Let A = {1, 2, 3} and B = {1, 2}, then $B \subset A$.

2. Improper subsets:

Let A be a subset of B. The set A is said to be an improper subset of B if A=B, then we write it as $A \subseteq B$.

Example 2: Let A = {1, 2, 3, 4} and B = {4, 3, 2, 1}.
Here B is an improper subset of A.

Example 3. Find all possible subsets of A = {x, y, z}.
All possible subsets are:
$\varnothing$, {x}, {y}, {z}, {x,y}, {x,z}, {y,z}, {x,y,z}.
Here Number of subsets = $2^3 = 8$.

Power set P(A) of a set A:

Power Set: Let A be a set. The set of all possible subsets of A is called the power set of A. It is denoted by P(A).

Example 4: Find the power set of A = {a, b}.

Solution: Here, number of subsets = $2^2 = 4$. All possible subsets are:
$\varnothing$, {a}, {b}, {a, b}. So, P(A) = {$\varnothing$, {a}, {b}, {a, b}}.

Note:
(i) Number of all possible subsets of an empty set is $1 = 2^0$
(ii) Number of all possible subsets of a set having one element is $2 = 2^1$
(iii) Number of all possible subsets of a set having two elements is $4 = 2^2$
(iv) Number of all possible subsets of a set having three elements is $8 = 2^3$

EXERCISE 1.2

1. Indicate which one is a subset of the other:
(i) A = {5, 6, 7}and B = {1, 2, 3, ..., 10} **(ii)** C = {2} and D={2, 3, 5}
(iii) T = {1, 2, 3, ..., 8} and S={2, 4, 6}

2. Find any three proper subsets of:
(i) A = {a, e, i, o, u} **(ii)** B = {x, y}

3. Find any two proper subsets and one improper subset of:
(I) X = {2, 4} **(ii)** Y = Set of prime numbers less than 4

4. Find all possible subsets of:
(i) F = {1, 2, 3} **(ii)** A = Set of even prime numbers
(iii) B = Set of odd prime numbers less than 7
(iv) C = {x | x $\in$Z, $-2 < x < 2$} **(v)** D = {x | x$\in$A}

5. Find the power set of:
(I) A = {1, 3, 5} **(ii)** B = {a, b, c}
(iii) C = $\varnothing$

6. **Find a set which has only one proper subset.**

7. **Find a set which has no proper subset at all.**

8. Find improper subsets of the following:

(i) $A = \{x \mid x \in O, x < 7\}$. **(ii)** $B = \{x \mid x \in P, x \leq 11\}$.

9. Let X = {20,30,40,50,60,80,100}. Find the number of elements in the following subsets of X:

(i) A = {Set of numbers divisible by 2}.
(ii) B = {Set of numbers divisible by 3}.
(iii) C = {Set of numbers divisible by 15}.
(iv) D = {Set of numbers divisible by 25}.

10. Decide which one is a subset of the other:

(i) N and W **(ii)** Z and N **(iii)** P and N
(iv) E and Z **(v)** Q and O **(vi)** Q and E

OPERATION ON SETS

Verify commutative and associative laws with respect to union and intersection.

(A) Commutative laws w.r.t union and intersection

Let A and B be any two sets.
The commutative law w.r.t union is:
- $A \cup B = B \cup A$

The commutative law w.r.t intersection is:
- $A \cap B = B \cap A$

Example 1: If $A = \{1, 2, 3\}$ and $B = \{1, 3, 5\}$, then verify commutative laws w.r.t union and intersection.

Solution: (i) Commutative law w.r.t union

We have to show that $A \cup B = B \cup A$

L.H.S: $A \cup B = \{1, 2, 3\} \cup \{1, 3, 5\}$
$= \{1, 2, 3, 5\}$

RHS: $B \cup A = \{1, 3, 5\} \cup \{1, 2, 3\}$
$= \{1, 2, 3, 5\}$

Thus,

LHS = RHS
$A \cup B = B \cup A$ **Hence verified.**

Solution: (ii) Commutative law w.r.t intersection:
We have to show that $A \cap B = B \cap A$.

LHS: $A \cap B = \{1, 2, 3\} \cap \{1, 3, 5\}$
$= \{1, 3\}$

RHS: $B \cap A = \{1, 3, 5\} \cap \{1, 2, 3\}$
$= \{1, 3\}$

Thus,
LHS = RHS **Hence verified.**

(B) Associative laws w.r.t union and intersection
Let A, B, and C be any three sets.
The associative law w.r.t union is: $A\cup(B\cup C) = (A\cup B)\cup C$
The associative law w.r.t intersection is: $A \cap (B \cap C) = (A \cap B) \cap C$

Example 2: If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$ then verify:
(I) $A \cap(B \cap C) = (A \cap B) \cap C$.
(ii) $A\cup(B\cup C) = (A\cup B)\cup C$.
Solution (I):
$A \cap (B \cap C) = (A \cap B) \cap C$ [Associative law w.r.t intersection]

L.H.S: $A \cap (B \cap C)$
$= \{1, 2, 3\} \cap (\{2, 3, 4\} \cap \{3, 4, 5\})$
$= \{1, 2, 3\} \cap \{3, 4\}$
$= \{3\}$

R.H.S: $(A \cap B) \cap C$
$= (\{1, 2, 3\} \cap \{2, 3, 4\}) \cap \{3, 4, 5\}$
$= \{2, 3\} \cap \{3, 4\}$
$= \{3\}$

Thus,
L.H.S = R.H.S, **Hence verified.**

Solution (ii):
$A\cup(B\cup C) = (A\cup B)\cup C$ [Associative law w.r.t union]

L.H.S: $A\cup(B\cup C)$
$= \{1, 2, 3\} \cup (\{2, 3, 4\} \cup \{3, 4, 5\})$
$= \{1, 2, 3\} \cup \{2, 3, 4, 5\}$
$= \{1, 2, 3, 4, 5\}$

R.H.S: $(A\cup B)\cup C$
$= (\{1, 2, 3\} \cup \{2, 3, 4\}) \cup \{3, 4, 5\}$
$= \{1, 2, 3, 4\} \cup \{3, 4, 5\}$
$= \{1, 2, 3, 4, 5\}$

Thus,
L.H.S = R.H.S, **Hence verified.**

To verify distributive laws:
Let A, B, and C be any three sets.
Then the distributive law of union over intersection is:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

And the distributive law of intersection over union is:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Example:
If $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$, and $C = \{2, 3, 6\}$, verify distributive laws.

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Solution (i):
L.H.S: $A \cup (B \cap C)$
$= \{1, 2, 3\} \cup (\{2, 4, 6\} \cap \{2, 3, 6\})$
$= \{1, 2, 3\} \cup \{2, 6\}$
$= \{1, 2, 3, 6\}$
R.H.S: $(A \cup B) \cap (A \cup C)$
$= (\{1, 2, 3\} \cup \{2, 4, 6\}) \cap (\{1, 2, 3\} \cup \{2, 3, 6\})$
$= \{1, 2, 3, 4, 6\} \cap \{1, 2, 3, 6\}$
$= \{1, 2, 3, 6\}$
Thus,
L.H.S = R.H.S **Hence verified.**

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Solution (ii):
L.H.S: $A \cap (B \cup C)$
$= \{1, 2, 3\} \cap (\{2, 4, 6\} \cup \{2, 3, 6\})$
$= \{1, 2, 3\} \cap \{2, 3, 4, 6\}$
$= \{2, 3\}$
R.H.S: $(A \cap B) \cup (A \cap C)$
$= (\{1, 2, 3\} \cap \{2, 4, 6\}) \cup (\{1, 2, 3\} \cap \{2, 3, 6\})$
$= \{2\} \cup \{2, 3\}$
$= \{2, 3\}$
Thus,
L.H.S = R.H.S **Hence verified.**

State and Verify De Morgan's Laws:
Let A and B be subsets of a universal set U. Then De Morgan's Laws are:

$(A \cup B)' = A' \cap B'$.
$(A \cap B)' = A' \cup B'$. **(Where dashes denote complements.)**

Example:
If U = {1, 2, 3, 4, 5, 6}, A = {1, 3, 5}, and B = {2, 4, 6}, verify any one of **De Morgan's Laws**.
Solution: One of De Morgan's Laws is:
$(A \cap B)' = A' \cup B'$
L.H.S:
Here, $A \cap B = \{1, 3, 5\} \cap \{2, 4, 6\}$
$= \{\ \}$
$(A \cap B)' = U - (A \cap B) = \{1, 2, 3, 4, 5, 6\} - \{\ \}$
$= \{1, 2, 3, 4, 5, 6\}$
R.H.S:
Here,
$A' = U - A = \{1, 2, 3, 4, 5, 6\} - \{1, 3, 5\}$
$= \{2, 4, 6\}$
$B' = U - B = \{1, 2, 3, 4, 5, 6\} - \{2, 4, 6\}$
$= \{1, 3, 5\}$
$A' \cup B' = \{2, 4, 6\} \cup \{1, 3, 5\}$
$= \{1, 2, 3, 4, 5, 6\}$
Thus,
L.H.S = R.H.S **Hence verified.**

EXERCISE 1.3

1. If A = {1, 2, 3, 4, 5}, B = {1, 3, 5, 7}, and C = {2, 4, 6, 8} then verify:
(i) Commutative law w.r.t intersection and union.
(ii) Associative law w.r.t union and intersection.
(iii) Distributive law of intersection over union.
(iv) Distributive law of union over intersection.

2. If P = {a, b, c, d, e}, Q = {c, e, f, g, h}, and R = {d, e, h, i, j} then verify that:
(i) $P \cup Q = Q \cup P$

(ii) $P \cap Q = Q \cap P$ **(iii)** $P \cup (Q \cup R) = (P \cup Q) \cup R$
(iv) $P \cap (Q \cap R) = (P \cap Q) \cap R$

3. If $U = \{1, 2, 3, ..., 10\}$, $A = \{1, 2, 4, 5\}$, and $B = \{2, 4, 6, 8, 10\}$ then verify that:
(i) $(A \cup B)' = A' \cap B'$. **(ii)** $(A \cap B)' = A' \cup B'$.

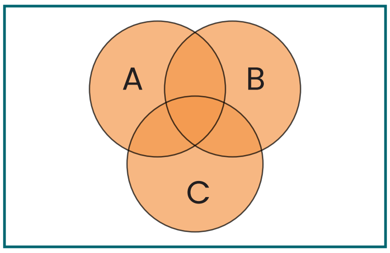
4. If $U = \{x \mid x \in O, x<11\}$, $Q = \{x \mid x \in P \wedge 3< x < 11\}$ and $R = \{x \mid x \in E \wedge x<10\}$, then verify distributive laws.

5. If $U = \{x \mid x \in N \wedge x <11\}$, $A = \{x \mid x \in E \wedge 2< x <10\}$ and $B = \{x \mid x \in P \wedge 3 < x < 11\}$, then verify De Morgan's Laws.

VENN DIAGRAM

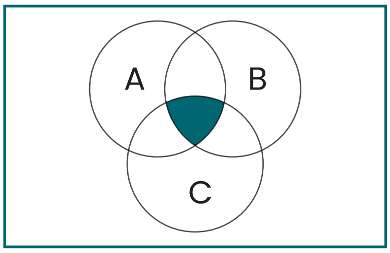
Demonstrate union and intersection of three overlapping sets through Venn Diagram:
We know that Venn diagram is a graphical representation of sets and their operations using geometrical shapes. Here Union and Intersection of three overlapping sets is shown through Venn diagram in the following figures:

Fig (ii): Union



Shaded portion represents Union

Fig (ii): Intersection



Shaded portion represents Intersection

To Verify Associative and Distributive Laws through Venn Diagram:

Associative and distributive laws can easily be verified through Venn diagrams as explained in the following examples.

Example:
If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $C = \{3, 4, 5\}$ then verify through Venn diagrams.
1. $A \cap (B \cap C) = (A \cap B) \cap C$.
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Solution:
(i) $A \cap (B \cap C) = (A \cap B) \cap C$
L.H.S.: $A \cap (B \cap C)$

Here **shaded portion represents** $B \cap C$

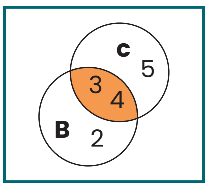


Fig. (i)

• **According to Fig (i):**
$B \cap C = \{3, 4\}$.

- **According to Fig (ii):**

$A \cap (B \cap C) = \{3\}$

Here **shaded portion represents** $A \cap (B \cap C)$.

R.H.S: $(A \cap B) \cap C$

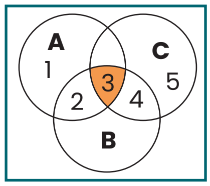


Fig. (ii)

- **According to Fig (iii):**

$A \cap B = \{2, 3\}$.

Here **shaded portion represents** $A \cap B$.

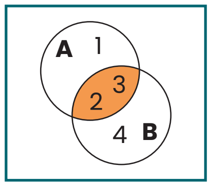


Fig. (iii)

- **According to Fig (iv):**

$(A \cap B) \cap C = \{3\}$.

Here **shaded portion represents** $(A \cap B) \cap C$.

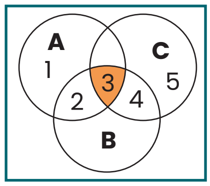


Fig. (iv)

Thus, L.H.S. = R.H.S.

Hence verified, the associative law w.r.t. intersection.

Solution: (ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

L.H.S. = $A \cup (B \cap C)$.

Here, **shaded portion represents** $B \cap C$.

- **According to Fig (v):**

$B \cap C = \{3, 4\}$.

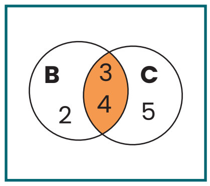


Fig. (v)

Activity: Verify the Associative Law w.r.t. union.

- **According to Fig (vi):** $A \cup (B \cap C) = \{1, 2, 3, 4\}$

R.H.S. = $(A \cup B) \cap (A \cup C)$

Here, **shaded portion represents** $A \cup (B \cap C)$.

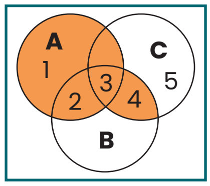


Fig. (vi)

- **According to Fig (vii):**

$A \cup B = \{1, 2, 3, 4\}$

Here, **shaded portion represents** $A \cup B$.

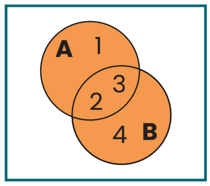


Fig. (vii)

- **According to Fig (viii):** $A \cup C = \{1, 2, 3, 4, 5\}$

Here, **shaded portion represents** $A \cup C$.

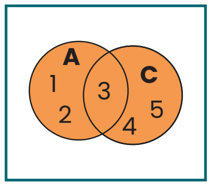


Fig. (viii)

- **According to Figure (ix):** $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4\}$.

Thus, L.H.S. = R.H.S.

Here, **shaded portion represents** $(A \cup B) \cap (A \cup C)$.

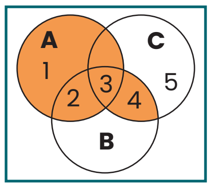


Fig. (ix)

$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. **Hence verified.**

The Distributive Law of Union over Intersection.

Activity: Students themselves verify Distributive Law of Intersection over Union through Venn Diagram.

EXERCISE 1.4

1. If A = {5, 6, 7}, B = {6, 7, 8}, C = {7, 8, 9}, then represent:
(I) $A\cup(B\cap C)$ **(ii)** $A\cap(B\cup C)$ through Venn diagrams.

2. If A = Set of even natural numbers less than 15.
B = Set of Prime numbers less than 15.
C = Set of Odd natural numbers less than 15.
then show the following by using Venn diagrams:
(I) $A\cup(B\cap C)$ **(ii)** $A\cap(B\cup C)$ **(iii)** $A\cup(B\cap C) = (A\cup B)\cap(A\cup C)$..

3. If A = {a, b, c, d}, B = {b, c, d, e} and C = { } verify through Venn diagram that:
(i) $A\cup(B\cup C) = (A\cup B)\cup C$ **(ii)** $A\cap(B\cup C) = (A\cap B)\cup(A\cap C)$
(iii) $A\cap(B\cap C) = (A\cap B)\cap C$

4. If D = {x|x ∈ N ∧ x<10}, E = {x|x ∈ W ∧ x <8}, F = {x|x ∈ P ∧ x < 11}
then verify by using Venn diagram:
(i) $D\cup(E\cap F) = (D\cup E)\cap(D\cup F)$. **(ii)** $D\cap(E\cup F) = (D\cap E)\cup(D\cap F)$.

5. Choose the correct answer:

(i) Which one is a subset of W?
(a) Z **(b)** N **(c)** Q **(d)** E

(ii) Number of all subsets of A={a, e, i, o, u} is:
(a) 8 **(b)** 16 **(c)** 32 **(d)** 64

(iii) For any three sets A,B, and C, $(A\cup B)\cap(A\cup C)$ = ________________.
(a) $A\cap(B\cup C)$ **(b)** $A\cup(B\cap C)$ **(c)** $A\cap(B\cap C)$ **(d)** None of these

(iv) In Venn diagram, universal set is usually denoted by:
(a) Triangle **(b)** Rectangle **(c)** Circle **(d)** Star

6. Find P(A) if A = {x, y, z}.

7. If A = {1, 2, 3}, B = {2, 4, 6}, C = {x|x ∈ W ∧ x < 4} then verify:
(i) $(A\cap B)\cap C = A\cap(B\cap C)$. **(ii)** $A\cup(B\cup C)=(A\cup B)\cup C$.
(iii) $(A\cup B)\cap C = (A\cap B)\cup(A\cap C)$.

8. Choose the correct answer.

(i) If A = {2, 4, 6, 8}, B = {1, 3, 5, 7, 9}, then A∩ B is:

(a) {2, 4} **(b)** { } **(c)** {1, 3} **(d)** {2, 9}

(ii) If P = {0, 2, 6}, Q = {1, 3} then P∪Q is:

(a) {2, 4} **(b)** {1, 3} **(c)** {0, 1, 3} **(d)** {0, 1, 2, 3, 6}

(iii) Number of subsets of power set of a set having four elements is:

(a) 8 **(b)** 16 **(c)** 64 **(d)** 32

(iv) U = {1, 2, 3, 4, 5}, A = {1, 3}, then A′ is:

(a) {2,4,5} **(b)** {1,2,3} **(c)** {4,5} **(d)** {1}